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PALEOMAGNETISM OF EARLY APHEBIAN DIABASE DYKES FROM THE SLAVE STRUCTURAL PROVINCE, CANADA*

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ABSTRACT

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Early Aphebian dykes (lowermost Proterozoic) intrude the Archean terrain of the Slave Structural Province of the Canadian Shield and paleomagnetic results from them are reported. The Dogrib dykes, with an Rb/Sr age of 2692 ± 80 m.y., have directions of magnetization directed toward the NW without reversals (16 sites; 309, +37; $\alpha_{95} = 4^{\circ}$; pole 35S, 050W). The Indin dykes, with an Rb/Sr age of 2093 ± 86 m.y., have magnetization directed toward the SE with reversals (13 sites; 131, +58; $\alpha_{95} = 8^{\circ}$; pole 19N, 076W). Other, less well-documented data from a third dyke swarm (the "X" dykes) and a basic sill (the Duck Lake Sill), are also presented, and a very tentative polar path for the Slave Province in the earlier Proterozoic is given. This path is not greatly different from a similar very tentative early Aphebian polar path from the Archean Superior Province, considering the uncertainties in the paleomagnetic and age determinations. We interpret this to mean either that the intervening Hudsonian Structural Province (—1850 m.y.) was not the site of a wide plate-style opening and closing ocean, or if it was, the two bounding Archean cratons returned approximately to their original relative position.

INTRODUCTION

There are two large blocks of Archean terrain (older than 2500 m.y.) in the Canadian Shield, the Superior and Slave Structural Provinces (Fig. 1). These blocks became stabilized at the end of the Kenoran orogeny at about —2700 m.y. (Stockwell, 1972). They are separated by the Churchill Structural Province which consists of Archean terrain with belts of younger rocks that were for the most part metamorphosed during the Hudsonian orogeny which ended at about —1850 m.y. (Stockwell, 1972). The Aphebian is the interval between the ends of the Kenoran and Hudsonian orogenies. There is a third

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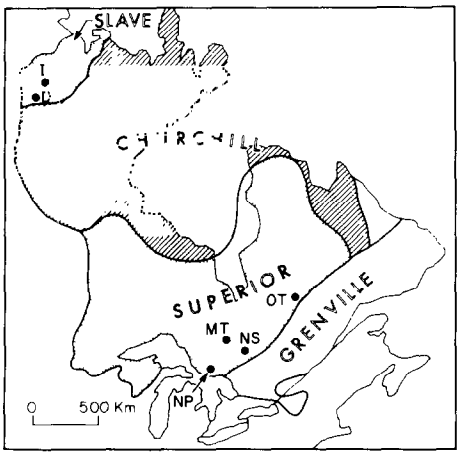


Fig. 1. Sketch-map showing the Archean Slave and Superior cratons bounding the Churchill Structural Province. *I* and *D* are the Indin and Dogrib dykes reported herein. The other paleomagnetic sampling localities are as follows: *OT* Otish Gabbro; *NS* and *NP* Nipissing Diabase from two different areas; *MT* Matachewan dykes.

Archean block — the Bear Tooth Mountain (BTM) uplift — in the U.S.A. If the Hudsonian orogen is the product of plate tectonics then it is to be expected that the bounding Archean cratons moved relative to one another, and these movements should be reflected in their paths of apparent polar movement. For example, if the Hudsonian orogen is the product of the Wilson cycle, that is, of an opening and closing ocean, then one possible paleomagnetic signature is a complimentary pair of pre-Hudsonian polar loops (each corresponding to movements of the bounding cratons during the opening and closing phases) which meet at their young end at the close of the Hudsonian orogeny, and which will come together, but not generally meet, at their older end (Fig. 2). Clearly if such a signature were observed, then it would be a good

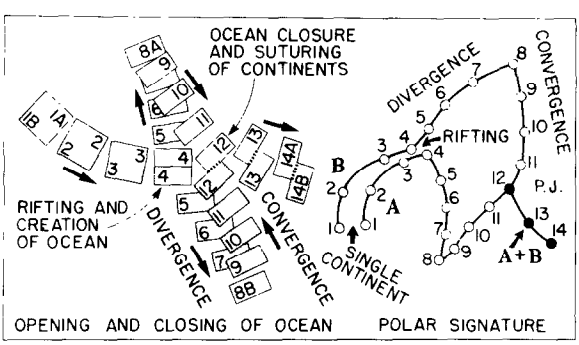


Fig. 2. Hypothetical polar signature of the Wilson cycle. *A* and *B* are two continental elements which diverge and then converge, returning to about the same positions relative to one another. One possible corresponding polar signature is given on the right.

TABLE I

Dyke nomenclature

Used herein	Burwash et al. (1963)
Mackenzie[1]	Mackenzie 3
Indin[2]	Mackenzie 2 + 4
"X"[3]	Mackenzie 1
Dogrib[4]	Mackenzie 1

[1] Fahrig and Wanless (1963). [2] [3] [4] herein.

argument for the plate-style origin of the Churchill Province. They do not meet exactly at their older end because the bounding cratons will not return to the same positions exactly. In this paper we present the first paleomagnetic data from pre-Hudsonian rocks of the Slave Province, for comparison with data from the other Archean cratons.

In the Yellowknife area in the Slave Structural Province lavas and graywacke—shale tubidites make up the Yellowknife Supergroup. They are metamorphosed up to amphibolite grade, folded, and cut by granitic intrusions which yield Rb/Sr whole-rock isochron ages ranging from 2500 m.y. to 2660 m.y. (Green and Baadsgaard, 1971), and K/Ar ages ranging from approximately 2300 to 2600 m.y. (Fahrig and Wanless, 1963). This terrain is intruded by several swarms of vertically dipping unmetamorphosed diabase dykes. In Table I the various swarms are listed by the names used in this paper, along with the earlier designations of Burwash et al. (1963). The Mackenzie dykes, which are about 1250 m.y. old, have been extensively studied previously (Robertson, 1969; Fahrig and Jones, 1969; Irving et al., 1972). In this paper we give paleomagnetic results from the three older sets that range in age from perhaps 2000 m.y. to as much as 2650 m.y.

Five to ten cores were drilled at each sampling site and two specimens cut from each core. Baked country rocks were sampled whenever possible. Alternating field (a.f.) demagnetization was carried out on all specimens. Stable-end point determinations were determined for one specimen from each core, and the second specimen was then demagnetized in that field. The results from each pair of specimens were then averaged to give the mean for a core and this was the basic statistical datum used. Results were rejected (9 diabase and 19 baked contact cores) if the difference between the directions of specimen pairs exceeded 50°.

DOGRIB DYKES

Near Yellowknife there is a prominent set of ENE-trending dykes (Fig. 3) that range in width from 10 to 150 m. The basement in this area is cut by N

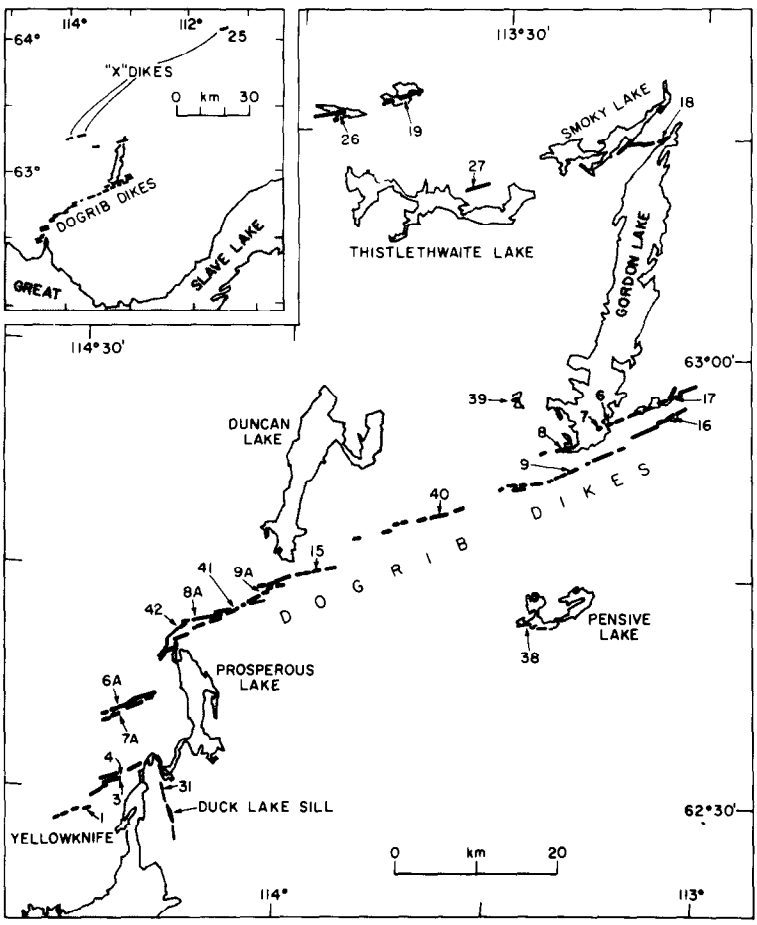


Fig. 3. Sampling localities of Dogrib and "X" dykes and the Duck Lake Sill. The numbers are field designations and have no geological significance.

to NE-trending faults and left-lateral displacement of the dykes occur along them. The dykes can be traced from Yellowknife to Gordon Lake, a distance of 100 km. They are named here the *Dogrib dykes* and have been sampled at sixteen sites. They are part of the Mackenzie Set 1 dyke swarm of Burwash et al. (1963). Also included in Set 1 are some easterly trending dykes north of Thistlethwaite Lake and around Lac De Gras (Fig. 3). These dykes have magnetization directions different from the Dogrib's and are referred to as the "X" dykes. In addition, several other intrusions were sampled, which at the time were thought to be part of Set 1, but were found to have other magnetization directions.

Dogrib diabase consists of almost equal portions of pyroxene (augite) and plagioclase with ophitic texture. There is about 1% olivine, and less than 1% of iron ores. The minerals are generally little altered. Pyroxenes are occa-

sionally partially altered to green hornblende and minor chlorite, and the feldspars to white micas. Olivine is always altered, often being completely changed to a felt of chlorite and other magnesian-rich minerals, with numerous small grains of magnetite. The iron ores consist essentially of blocky grains consisting mostly of maghemite with lamellae of ilmenite, and a few discrete grains of ilmenite. The arrangement of the ilmenite lamellae suggests that their host mineral was originally magnetite that was almost completely converted to maghemite, and locally to hematite. Residual magnetite occurs as blebs in maghemite. Sulfides occur in very minor amounts.

Country rock was sampled at ten localities between 0 and 10 m from dyke margins. Their iron minerals content is very much less than 1%. In the baked Yellowknife sedimentary rocks a few small laths of ilmenite, partially altered to leucoxene occurred along cleavages in biotite grains, and minor sulphides occur. In the baked granitic rocks, magnetite and ilmenite occur as fine blocky grains.

Twenty-one samples from the Dogrib dykes have been dated by the whole-rock K/Ar method by Leech (1966) and gave ages ranging from 900 m.y. to 2310 m.y. A profile of seven samples across one dyke yielded ages from 900 to 1930 m.y. Leech's preferred age for Mackenzie Set 1 dykes, of which the Dogrib dykes are a part, was 2200 m.y. From K/Ar studies, Fahrig and Wanless (1963) suggested an age of about 2150 m.y. More recently these dykes have been dated by Rb/Sr whole-rock method and an age of 2692 ± 80 m.y. has been obtained (Gates and Hurley, 1973).

Sampling sites are shown in Fig. 3. Typical a.f. demagnetization curves are given in Fig. 4. Median destructive fields (m.d.f.) are usually between 50 and

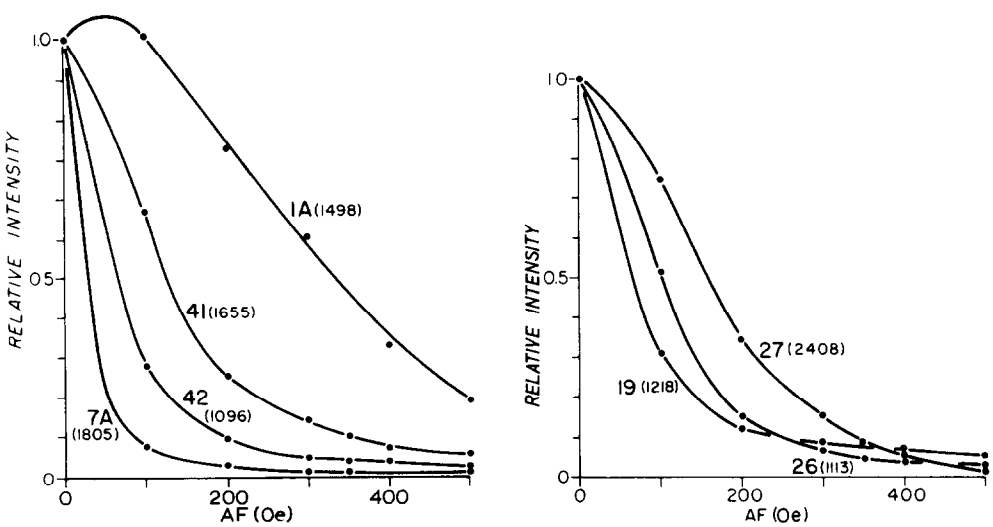


Fig. 4. Typical a.f. demagnetization curves from the Dogrib (left) and "X" (right) dykes. The n.r.m. intensities in microgauss are given in brackets.

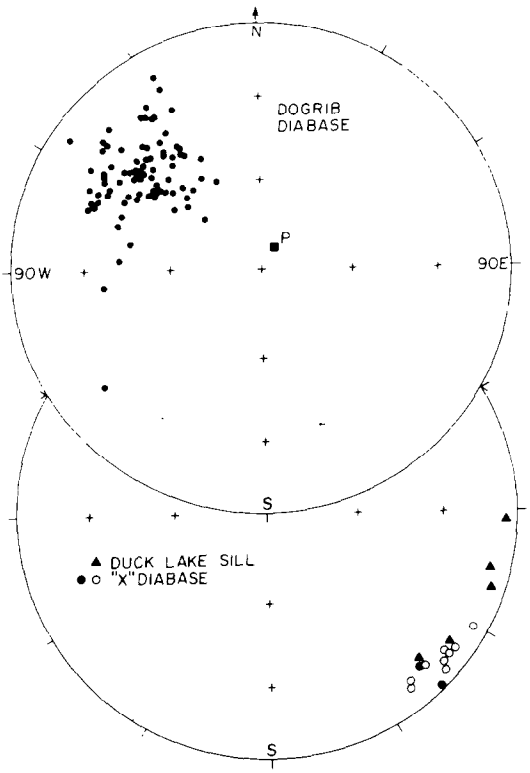


Fig. 5. Directions of magnetization in the Dogrib and "X" dykes and in the Duck Lake Sill after a.f. cleaning. Open symbols denote upward inclination. *P* is the present field. The primitive is the present horizontal. All core directions (mean of two specimens) are plotted.

150 Oe. The directions of natural remanent magnetization (n.r.m.) are scattered widely about the present field. N.r.m. intensities range from 110.0 to $0.04 \cdot 10^{-3}$ gauss. During demagnetization the directions of most specimens migrate toward the WNW. The directions are often not greatly changed in fields of 500 and 750 Oe, but in higher fields become random. The cleaned directions for all cores are plotted in Fig. 5, and the mean site directions listed in Table II. At four sites (8A, 9A, 15, 16) specimens from near the margins of thick dykes have end-points with directions toward the SE (see below).

Samples of Yellowknife sediment and granitic rocks near diabase contacts are either too weak to measure (less than $0.5 \cdot 10^{-6}$ gauss) or became random in direction in alternating fields of less than 100 Oe, probably reflecting their low iron oxide content (see above). Only twelve samples had any directional stability. Seven are grouped about the present field, and five have SE directions. Hence there is no direct evidence that the stable magnetization in the dykes was acquired at the time of cooling following intrusion.

TABLE II

Dogrib dyke results

No.	A.f. (Oe)	N	D, I (R)
1	2—4	10	313, +43 (9.88)
3	2—4	5	326, +30 (4.89)
6	3—4	5	320, +40 (4.97)
6A	4	7	312, +40 (6.77)
7	3	5	317, +41 (4.96)
7A	4	3	397, +37 (2.94)
8	3	3	316, +40 (2.92)
8A	3—4	9	305, +48 (8.80)
9	4	5	306, +31 (4.95)
9A	3—4	4	317, +41 (3.87)
15	4	4	301, +39 (3.82)
16	3	3	309, +35 (3.00)
17	4	5	302, +26 (4.11)
40	4	6	311, +27 (5.91)
41	4	5	292, +29 (5.00)
42	3	5	302, +35 (4.96)

Site numbers are given in the first column. A.f. is the cleaning field in hundreds of oersted (peak). N is the number of cores, two specimens measured from each. D, I is the mean site direction giving cores unit weight. R is the resultant.

TABLE III

Directions during thermal demagnetization of stable remanence

°C	Site 1	Site 40
20	308, +45	307, +22
100	308, +44	304, +22
200	315, +36	307, +26
300	312, +31	311, +19
400	310, +33	307, +25
500	305, +39	278, +48
550	303, +08	295, +09
600	321, +35	300, +25

The specimens were first demagnetized in a.f. of 400 Oe to remove unstable components, and then thermally demagnetized. The n.r.m. intensities were 1298 and 1705 microgauss, respectively. After a.f. demagnetization these reduced to 104 and 71, respectively, and it is this stable magnetization which gives the above results.

TABLE IV

Summary of results

	n	D, I	k	α_{95}	Pole	A_{95}	dm, dp
Indin NE	5	134, +51	17	19	—	—	—
Indin NW	8	128, +62	64	7	—	—	—
Indin total (63N, —114W)	13	131, +58	29	8	19N, 076W	10	12,9
Indin total (c)	10	135, +58	29	9	18N, 079W	11	13,10
"X" (63.3N, 113.6W)	3	132, —05	100	12	20S, 061W	—	12,6
Dogrib (62.8N, 113.8W)	16	309, +37	70	4	35S, 050W	4	5,3

n is the number of sites; D, I the mean direction; k the precision; α_{95} Fisher's circle of confidence around the direction; A_{95} the circle of confidence around the pole calculated from site poles; and dm, dp the semi-axes of the elliptical polar error.

In order to determine the blocking temperatures of the harder magnetic component in the diabase, thermal demagnetization was carried out on specimens from sites 1 and 40 after demagnetization in 400 Oe. The averaged directions, however, remain essentially unchanged up to 600°C (Table III). This indicates that the hard magnetization is due to titanium-free magnetite and hematite and both have essentially the same direction. Presumably these are the minor magnetite and hematite observed in polished sections. There is no large decrease in intensity between 300 and 400°C (the temperature at which maghemite inverts to hematite), as would be expected if the stable remanence were due to maghemite which is the dominant phase observed in polished section. If we are correct in ascribing the magnetite to relicts of original crystals now largely altered to maghemite, and the hematite to oxidation, then this oxidation probably occurred soon after intrusion, probably during initial cooling, otherwise the magnetite and hematite components would not now have the same direction.

The summary statistics are given in Table IV. The between-site precision (which is significant) is high and presumably reflects to some extent the small number of independent bodies sampled.

"X" DYKES

Three dykes formerly thought to belong to the same swarm as the Dogrib dykes have very different directions (Fig. 5). These are referred to as the "X" dykes. They are more altered than the Dogrib dykes. Pyroxene is partially or completely altered to amphibole and chlorite, and plagioclase to white micas. The rocks contain minor amounts of quartz, and olivine, if present originally, has been completely altered. The iron ores consist of blocky grains of what was probably originally magnetite, partially or completely converted to maghemite, and minor amounts of hematite. Lamellae of ilmenite are abun-

TABLE V

Miscellaneous results

Location (lat. N, long. W)	N	A.f. ($Oe \times 10^2$)	D, I	R	α_{95}	Pole
19 "X" dike (63.3, 113.7)	4	3—4	133, 00	3.95	11	18S, 63W
25 "X" dike (64.5, 111.4)	3	3—4	125, —07	2.99	7	18S, 54W
26 "X" dike (63.3, 113.6)	4	3—4	137, —10	3.99	6	24S, 65W
27 Thistlethwaite dike (63.2, 113.6)	5	3	281, +72	4.99	3	53N, 175W
31 Duck Lake sill (62.5, 114.3)	5	3	113, +09	4.81	17	06S, 47W
38 Pensive Lake sheet (62.7, 113.4)	5	4	355, +63	4.99	4	72N, 077E
SE directions 8A, 9A, 15, 16, 19, 25, 26, 38	13	1—4	139, +47	12.43	9	07N, 78W
Yellowknife sediments 26 (63.3, 113.6)	3	n.r.m.	088, —75	2.99	6	52S, 018E

dant and discrete grains of ilmenite are common. The "X" dykes appear to have more ilmenite than do the Dogribs. Ilmenite is occasionally slightly altered to leucoxene. Sulphides are very minor constituents.

The results are summarized in Table V. The pole is in central Brazil. A few samples from each of these three dykes have SE directions. The a.f. decay curves are similar to those for the Dogrib dykes (Fig. 4). The initial intensities range from 0.02 to 4.9 · 10^{-3} gauss. The work of Leech (1966) and Fahrig and Wanless (1963) would suggest that the dykes are about the same age as the Dogrib.

INDIN DYKES

Burwash et al. (1963) recognized a set of NE and NW-trending dykes that are younger than the Dogrib dykes which they referred to as Mackenzie Sets 2 and 4 (Table I). Leech (1966), on the basis of K/Ar dates, suggested that both were the same age and formed a conjugate set. This has been confirmed by Gates and Hurley. Dykes from both trends have similar directions of magnetizations (see below). We therefore refer to them by a single name, the *Indin dykes*, as they are well-developed around Indin Lake.

Indin diabase is massive, displays ophitic texture, and is composed of about equal amounts of plagioclase and pyroxene, with minor olivine, iron ores, and sphene. The pyroxene is very slightly altered to hornblende and chlorite. Plagioclase is fresh, except for zoisite in the cores of many grains. Olivine is mostly altered to a felt of chlorite and hydrous silicates, magnetite, hematite, and goethite. The opaque minerals include blocky grains of magnetite, ilmenite, hematite, and minor sulphides. Magnetite commonly contains lamellae of ilmenite and in part is altered to fine intergrowths of hematite and ilmenite. Maghemite was not identified. Ilmenite is also present as discrete grains. It is always partially altered to leucoxene. Hematite occurs as fine grains in the silicate minerals.

K/Ar ages from the NW-trending dykes in the Yellowknife area range from 1205 to 1953 m.y., and the NE-trending dykes range from 1155 to 1965 m.y. (Leech, 1966). Leech's preferred mean age for both trends is about 2000 m.y. Rb/Sr whole-rock isochrons from the Indin dykes yield an age of 2093 ± 86 m.y. The isochrons for the NE and NW sets are not significantly different (Gates and Hurley, 1973). This isochron is made up of samples from the Yellowknife area and possibly the Lac de Gras area east of Yellowknife.

Sampling sites are shown in Fig. 6, and typical a.f. demagnetization curves in Fig. 7. The n.r.m. directions are widely scattered. The initial intensities range from 0.01 to 183 · 10^{-3} gauss. During demagnetization the direction migrates toward the SE with positive inclination or NW with negative inclination, falling into reversed and normal groups (Fig. 8). Only two specimens (marked "A" in Fig. 8) do not show this behaviour. The mean site directions are given in Table VI, and the summary statistics in Table IV. The NE and NW sets have directions that are not significantly different. The reversed

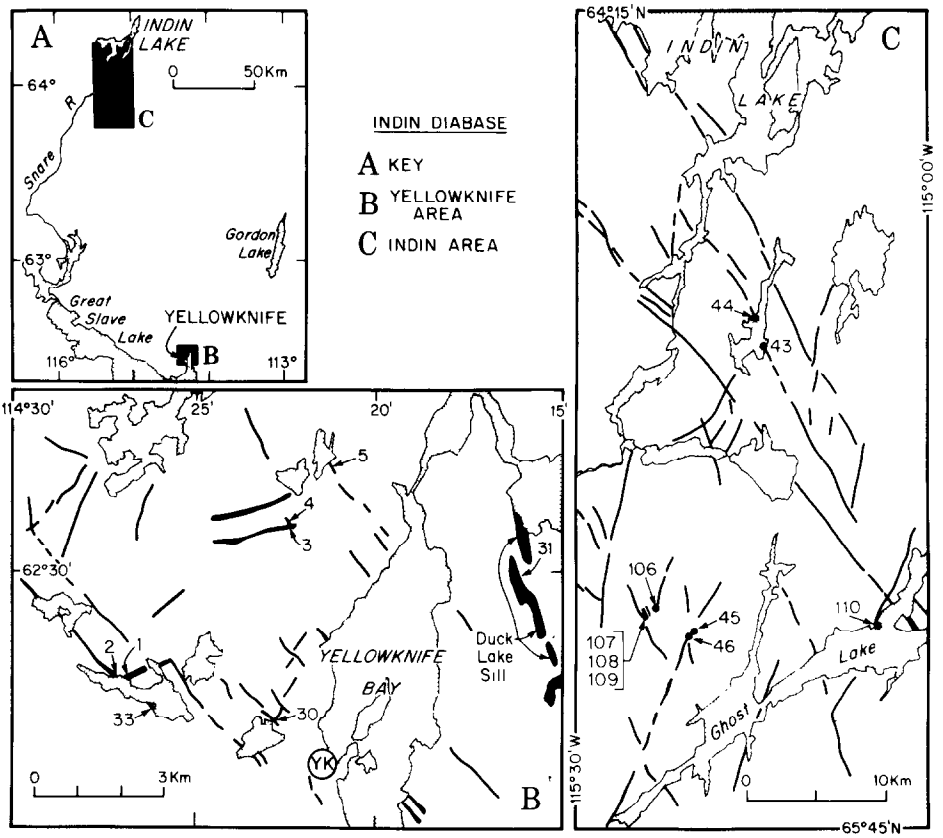


Fig. 6. Sampling localities of Indin dykes. Fig. 6A shows the two sampling areas around Indin Lake (Fig. 6C) and Yellowknife (Fig. 6B). The town of Yellowknife is marked *YK*. The numbers are field-collecting numbers and have no geological significance.

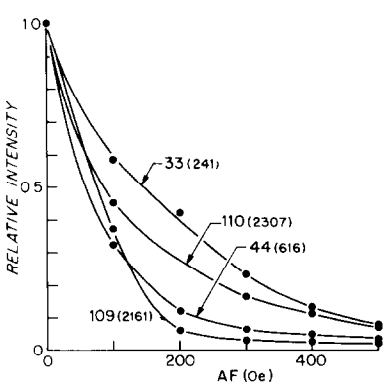


Fig. 7. Typical a.f. demagnetization curves of Indin diabase. N.r.m. intensities in microgauss are given in brackets.

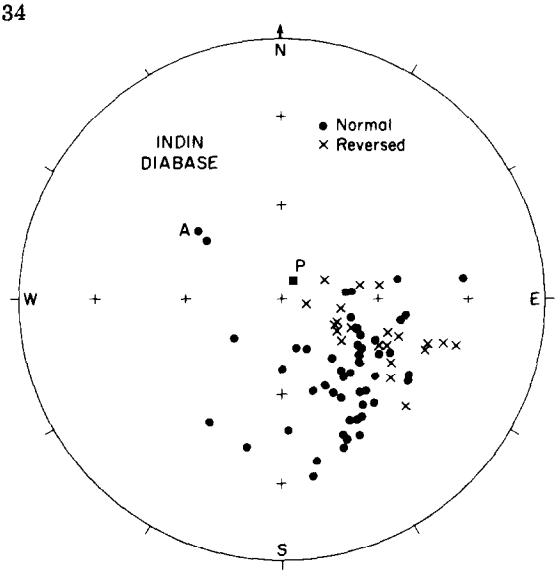


Fig. 8. Directions of magnetization of the Indin diabase. All directions are plotted on the lower hemisphere, the crosses are south-seeking directions, the solid dots north seeking. *P* is the present field. *A* are anomalous directions (see text). The primitive is the present horizontal.

TABLE VI

Indin dike results

No.	T	*N*	A.f. (Oe)	*D*, *I* (*R*)
002	NW	6	3	292, −66 (5.92)
005	NW	3	2–3	151, +65 (2.97)
030	NW	5	2–3	155, +61 (4.63)
030c	NW	4	1–2	152, +69 (3.83)
030	NE	6	3	179, +50 (5.79)
030c	NE	2	3–4	233, +36 —
033	NW	6	3	289, −71 (5.91)
033c	NW	4	2	305, −49 (3.98)
043	NW	6	3	132, +53 (5.87)
044	NW	6	1.5–5	111, +65 (5.92)
044′	NW	2	2	309, +58 —
045	NE	5	1–2	295, −45 (4.92)
046	NE	5	1–2	294, −47 (4.86)
106	NE	4	2–3	114, +55 (3.92)
107	NW	7	2–3	122, +58 (6.47)
109	NW	5	2–4	126, +54 (4.94)
110	NE	6	3–4	152, +43 (5.98)

Symbols as for Table IV. T is the dyke trend. At site 30 two dykes were sampled one trending NW and the other NE. Sites 44 and 44′ are the same dyke, the directions at 44′ are the aberrant direction shown in Fig. 8.

(n = 4; 293°, −57°; α_{95} = 15°) and normal (n = 9; 139°, +58°; α_{95} = 9°) sets are not markedly different irrespective of sign. Sites 45 and 46, 2 and 33, and 43 and 44, are from what appears to be the same dykes, and it may be fairer to count these as two rather than as four separate results. The average (n = 10, c in Table IV) calculated in this way is statistically indistinguishable from that giving each site unit weight (n = 13).

Samples were collected from near contacts at six sites. Stable directions were found at only three sites (labelled "C" in Table VI). At site 30NW the directions agree exactly with those in the dykes. At sites 30NE and 33 the inclinations are lower. The approximate, but inexact, agreement between dykes and their contact rocks is a common effect, and is presumably caused by interaction between the fields of contact and diabase as they become magnetized at somewhat different times (Irving et al., 1972; Park, in press). At greater distances from the Indin dykes, notably in Dogrib diabase around Yellowknife, the directions are very different. Hence the contact test shows that the magnetization of the Indin dykes was probably acquired at the time of cooling. Any pervasive later magnetization event would also have remagnetized the Dogrib diabase in the same direction, which it has not.

OTHER BODIES

Four other results, all showing high internal consistency, were obtained in the course of this work, and are reported here. They were collected in the first place because it was felt that like the "X" dykes they could belong to the Dogrib swarm. Evidently this is not so, and several, perhaps as many as four phases of diabase intrusion may be present in the original Mackenzie Set 1. The diabase dyke at Thistlethwaite Lake (site 27) and Yellowknife sediments at 20 m from diabase dyke at site 26 have antiparallel directions. A diabase sheet at Pensive Lake (38) is very stable. The Duck Lake Sill gave directions and pole rather close to those of the "X" dykes (Fig. 5). Five samples of the Duck Lake Sill have been dated by Leech (1966) by the K/Ar method and ages range from 1490 to 2090 m.y. She suggests that the intrusion is 2000—2100 m.y. old. No ages are available for the dykes. One site (18) yielded no stable magnetization. It is notable that the YK sediments at site 26 have directions very different from those in dyke 26 itself.

SE DIRECTIONS

Southeast directions with intermediate positive inclinations are found sporadically in diabase specimens from near the margins of some Dogrib dykes, in some Yellowknife sediments near diabase contacts, and in a few samples of "X" dykes. The pole for these SE directions (Table V) falls near that for the Indin diabase and those from rocks of 1700—1800 m.y. It is also close to the two poles derived from secondary components in the Nonacho (13N, 086W; McGlynn et al., 1974) and Kahochella (18N, 074W; McMurray et al., 1973)

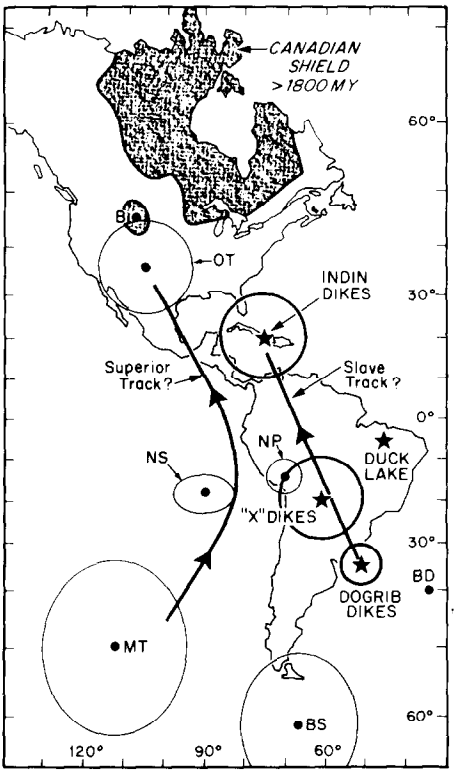


Fig. 9. Early Aphebian poles. The sampling locations are shown in Fig. 1. The results from the Slave Province are documented in Table IV. The results from the Superior Province are as follows: *OT* Otish Gabbro (Fahrig and Chown, 1973); *NP* and *NS* Nipissing diabase two results (Symons, 1970; Patel and Palmer, 1974); *MT* Matachewan diabase (Strangway, 1964; Fahrig et al., 1965). *B* is a pole from one unrecrystallized dyke from Beartooth Mountains (2550 m.y.; Larson et al., 1973).

sediments. This fact, and the sporadic occurrence of this magnetization, suggest that this magnetization was acquired during a period of crustal heating at about 1700 m.y.

DISCUSSION

The pole positions from the Slave Province are plotted in Fig. 9 and connected by a *very* tentative polar track. The age of this track on present evidence appears to be *about* 2600—2000 m.y. Undoubtedly future work will show that the pole path in this interval of time is very much more complicated, but it is of interest to make a tentative comparison with a polar path from the Superior Province based on three results, namely, the Matachewan diabase (2700 ± 200 m.y.; Gates and Hurley, 1973), the Nipissing diabase (2150 m.y.;

Van Schmus, 1965) and the Otish Gabbro (about 2000 m.y.; Fahrig and Chown, 1973). These polar reconstructions are necessarily very tentative indeed. Their salient feature is that although they may differ somewhat, they are reasonably close together. Thus, during the early Proterozoic, the Slave and the Superior Provinces could have been in *very approximately* the same relative positions as they are today. The results tell us nothing directly about relative movement subsequent to —2000 m.y., but they imply that *if* it was large then the Slave and Superior Provinces returned *approximately* to the same positions. The similarity is such as would be expected from the Wilson cycle, that is the pre-orogenic polar curves from bounding cratons agree approximately, but not exactly.

There are now numerous data from rocks in the age range 1700—1900 m.y. and their poles fall near to the present-day Carribean close to that for the older Indin dykes (McGlynn et al., 1974). Whether this means that the position of Laurentia relative to the pole has remained essentially unchanged for 300 m.y. during the later Aphebian or whether it is due to the magnetic updating, which has been shown to have occurred in some instances (McGlynn et al., 1974), remains to be determined.

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